

Track 2D Mean-Offset Full-Matrix Audit

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report extends the `Track 2` mean-centered collage diagnostic to the full official direction-valid candidate matrix. It computes raw curve error, curve-bias / `DC` offset, centered-shape error, peak-to-peak amplitude error, harmonic amplitude error, and harmonic phase error after every candidate produces predictions through its normal causal input path.

This is an analysis-only diagnostic. It does not train models, alter the dataset structure, update registries, or make full-curve mean-centering a deployment-time correction.

- Run Instance: `2026-06-03-10-54-10__track2d_mean_offset_full_matrix_audit`
- Config Path:
`config\paper_reimplementation\rcim_ml_compensation\reference_family_vs_feedforward\full_track2_matrix_template.yaml`
- Output Directory: `output\validation_checks\track2d_mean_offset_full_matrix_audit\2026-06-03-10-54-10__track2d_mean_offset_full_matrix_audit`
- Candidate Count: `111`
- Evaluated Curve Count: `12416`
- Harmonic Orders: `1, 3, 39, 40, 78, 81, 156, 162, 240`

Method

- raw `MAE` and `RMSE` are computed on the normal prediction residual;
- offset is the absolute difference between predicted and measured curve means;
- centered metrics subtract each curve's own mean from truth and prediction separately after inference;
- amplitude error compares predicted and measured peak-to-peak TE;
- harmonic amplitude and phase diagnostics are computed on selected sparse `RCIM` harmonic orders;
- condition summaries stratify by direction, speed, torque, and oil temperature.

Diagnostic Label Counts

Label	Candidate Count
shape	3
mixed:shape+amp	10
mixed:shape+amp+phase	3
mixed:shape+amp+phase+regime	2
mixed:shape+phase	4
mixed:shape+phase+regime	1
mixed:offset+amp	3
mixed:offset+amp+phase	7
mixed:offset+regime	26
mixed:offset+phase	15
mixed:offset+phase+regime	1
offset	36

Track 2D Diagnostic Ranking

Rank	Candidate	Surface	Raw MAE	Centered MAE	Offset	Gain [%]	Label
1	rcim_retuned_GBM19_Fw	Fw	0.001089	0.000832	0.000634	21.8	shape
2	rcim_original_ERT19_Fw	Fw	0.001471	0.000831	0.001138	38.6	offset
3	rcim_retuned_RF19_Fw	Fw	0.001487	0.000834	0.001183	40.0	offset
4	paper_retuned_best_Fw	Fw	0.001839	0.000833	0.001562	43.2	mixed:offset+regime
5	rcim_retuned_ERT19_Fw	Fw	0.001807	0.000833	0.001543	46.9	offset
6	rcim_original_RF19_Fw	Fw	0.001767	0.000837	0.001502	45.9	offset
7	rcim_original_DT19_Fw	Fw	0.001919	0.000834	0.001590	43.2	offset
8	rcim_retuned_DT19_Fw	Fw	0.001919	0.000834	0.001590	43.2	offset
9	rcim_original_LGBM19_Fw	Fw	0.001801	0.000846	0.001487	40.8	mixed:offset+regime
10	rcim_original_GBM19_Fw	Fw	0.001921	0.000834	0.001672	49.1	offset
11	rcim_retuned_ET19_Fw	Fw	0.002001	0.000835	0.001704	43.7	offset
12	rcim_retuned_LGBM19_Fw	Fw	0.001851	0.000846	0.001562	43.0	mixed:offset+regime
13	rcim_retuned_HGBM19_Fw	Fw	0.001851	0.000849	0.001554	42.7	mixed:offset+regime
14	ERT19_Fw	Fw	0.002295	0.000842	0.002062	52.6	mixed:offset+regime
15	rcim_original_HGBM19_Fw	Fw	0.002011	0.000859	0.001737	45.9	mixed:offset+regime
16	paper_original_best_Fw	Fw	0.002769	0.000833	0.002552	54.4	offset
17	rcim_original_ET19_Fw	Fw	0.002232	0.000896	0.001880	46.4	mixed:offset+regime
18	RF19_Fw	Fw	0.002164	0.000889	0.001844	45.5	mixed:offset+regime
19	rcim_retuned_XGBM19_Fw	Fw	0.002054	0.000852	0.001835	48.2	mixed:offset+regime
20	track1_best_Fw	Fw	0.003014	0.000857	0.002894	65.1	offset

Surface Leaders

Rank	Candidate	Surface	Raw MAE	Centered MAE	Offset	Gain [%]	Label
1	DT19_Bw	Bw	0.005226	0.000882	0.004999	55.1	mixed:offset+regime
1	rcim_retuned_GBM19_Fw	Fw	0.001089	0.000832	0.000634	21.8	shape
1	periodic_gru_sequence_global	global	0.002704	0.000992	0.002501	50.9	offset

Largest Mean-Offset Improvements

Rank	Candidate	Surface	Raw MAE	Centered MAE	Offset	Gain [%]	Label
94	harmonic_regression_global	global	0.018160	0.001758	0.018035	82.3	mixed:offset+amp+phase
43	LGBM19_Fw	Fw	0.006812	0.001098	0.006679	73.5	offset
48	XGBM19_Bw	Bw	0.007991	0.001191	0.007742	71.2	mixed:offset+regime
53	ELM19_Fw	Fw	0.007281	0.001437	0.007080	70.0	offset
41	HGBM19_Bw	Bw	0.006619	0.001112	0.006401	68.8	mixed:offset+regime
91	ELM19_Bw	Bw	0.010071	0.002022	0.009682	66.5	mixed:offset+regime
52	rcim_retuned_ELM19_Fw	Fw	0.007182	0.001435	0.006948	66.4	mixed:offset+regime
92	rcim_retuned_XGBM19_Bw	Bw	0.010679	0.002800	0.010116	65.9	mixed:offset+phase
20	track1_best_Fw	Fw	0.003014	0.000857	0.002894	65.1	offset
35	XGBM19_Fw	Fw	0.004190	0.000998	0.003987	63.2	mixed:offset+regime
46	ET19_Bw	Bw	0.006273	0.001387	0.005763	62.2	mixed:offset+regime
22	DT19_Fw	Fw	0.003122	0.000872	0.002972	61.9	offset
50	rcim_original_ELM19_Fw	Fw	0.005423	0.001461	0.005039	61.6	mixed:offset+regime
51	SVM19_Bw	Bw	0.004822	0.001544	0.004261	61.1	mixed:offset+regime
37	SVM19_Fw	Fw	0.003236	0.001175	0.002894	61.0	offset

Interpretation

The ranking is a diagnostic ordering, not a promotion rule. Offset-limited candidates need a causal offset-calibration or offset-aware loss strategy. Centered-shape-limited candidates need waveform-shape, derivative, harmonic amplitude, or phase improvements before retraining should be expanded.

The next training decision should keep `Fw`, `Bw`, and `global` surfaces in parallel. A forward leader does not close the backward or global branch, and a backward leader does not close the forward or global branch.

Runtime Input Boundary

All full-curve operations are post-prediction diagnostics. Candidate models still consume only current point-level state, an explicitly supported short causal history, or causal derived features. The audit does not use future TE samples as model inputs.

Machine-Readable Artifacts

- `output\validation_checks\track2d_mean_offset_full_matrix_audit\2026-06-03-10-54-10__track2d_mean_offset_full_matrix_audit\track2d_per_curve_metrics.csv`
- `output\validation_checks\track2d_mean_offset_full_matrix_audit\2026-06-03-10-54-10__track2d_mean_offset_full_matrix_audit\track2d_candidate_summary.csv`
- `output\validation_checks\track2d_mean_offset_full_matrix_audit\2026-06-03-10-54-10__track2d_mean_offset_full_matrix_audit\track2d_surface_leaderboard.csv`
- `output\validation_checks\track2d_mean_offset_full_matrix_audit\2026-06-03-10-54-10__track2d_mean_offset_full_matrix_audit\track2d_condition_stratified_summary.csv`
- `output\validation_checks\track2d_mean_offset_full_matrix_audit\2026-06-03-10-54-10__track2d_mean_offset_full_matrix_audit\track2d_mean_offset_full_matrix_audit_summary.yaml`